

CHINA'S GLOBAL INFLUENCE: A SURVEY THROUGH THE LENS OF INTERNATIONAL TRADE

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Abstract. One of the most impressive changes in the global economy in the past half-century has been China's high and sustained growth and its integration into the global economy. This phenomenal change not only brings huge benefits to the Chinese people, but also exerts a tremendous influence on the rest of the world. How large is China's influence on the global economy? How are various countries/regions affected differently? How are various industries in a country affected differently? Answering these questions is not simple. In this paper, we review the recent literature that attempts to answer these questions. The findings of the survey conducted in this study aim to provide a complete picture of China's global influence and to identify issues that require further analysis.

1. INTRODUCTION

China began its economic reform and adopted an open-door policy in 1979. This development has brought two significant changes to China: (i) rapid growth and development of the domestic economy; and (ii) graduate integration of the domestic economy into the global economy. China's GDP increased from approximately US\$201bn in 1979 to US\$4864bn in 2013, with an average annual growth rate of roughly 9.83%. China's exports increased from US\$30bn in 1979 to US\$2033bn in 2013, with an average annual growth rate of approximately 13.18%. Moreover, imports increased from US\$29bn in 1979 to US\$1696bn in 2013, with an average annual growth rate of roughly 12.71%.¹ In 1979, China's total trade ranked 32nd in the world. In 2014, China became the largest trading country in goods trade and the second largest in goods and services combined, after the United States.

To a large extent, China's open-door policy resembles the switch from autarky to free trade, as illustrated in standard international trade textbooks. Standard trade theories predict that China's economic growth and trade expansion bring both opportunities and challenges to the rest of the world. As for opportunities, on the one hand, given that the rapid economic growth has made China the second largest economy in the world, its huge market creates strong demands for imports of both raw materials and intermediate goods for infrastructure building and manufacturing production, as well as for imports of final goods consumption. China has also become one of the most attractive places for foreign direct investments. On the other hand, China supplies a vast amount of cheap manufacturing goods, which benefits consumers in the importing coun-

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¹ Data are from the World Bank with constant 2005 US dollar price.

tries as they can enjoy a wide variety and an extensive quantity of products at low prices. Moreover, the integration of China into global production chains helps enhance the efficiency of global production. As for challenges, countries that share the comparative advantages of China might suffer from severe competition in both domestic and third markets. Industries that are in competition with China might shrink and some firms might even exit the market.

Although the integration of China into the global market may not offer any new theory, its sheer size attracts people's attention because of its potentially huge influence on the rest of the world. This paper critically reviews the literature on the influence of China on the rest of the world with a focus on the channel of international trade. China has an abundant labour supply, rich resources, high innovative capability, a vast domestic market and very unbalanced levels of regional development. Moreover, it has a huge GDP but low per capita income. Consequently, its trade exerts different effects on various countries and industries. We initially examine the effects on developed countries (Section 2), then Asian countries (Section 3), then other developing countries (Section 4) and, finally, the world as a whole (Section 5).

2. DEVELOPED COUNTRIES

The effect of competition from low wage countries (LWC) on developed countries has been a hot topic in recent years. Among LWC, China stands out. In this section, we first review the debate about the effect of China's competition on the labour markets in developed countries. We subsequently discuss the mechanisms and evidence of trade-induced resource reallocation, productivity growth and technological progress in developed countries. Finally, we evaluate the claim of China exporting deflation.

2.1. *Labour market*

How does trade affect income inequality in developed countries? This question is old, but it has received renewed attention recently primarily because of China's deepening integration into the global economy. Researchers are particularly interested in China's effects on developed countries with regard to unemployment, the wage gap and labour participation. In this subsection, we initially focus on the US market and then on other developed countries.

2.1.1. *United States*

In a widely cited paper based on data from 1990 to 2007, Autor *et al.* (2013a) rate every US county for its manufacturers' exposure to competition from China. They determine that regions most exposed to China tend not only to lose more manufacturing jobs, but also see an overall employment decline and reduced rate of labour force participation. Regions with a higher exposure also have a larger increase in the number of workers receiving unemployment insurance, food stamps and disability payments. Specifically, Autor *et al.* demonstrate

that import competition explains one-quarter of the contemporaneous aggregate decline in US manufacturing employment. However, they do not find any significant effect on the wage gap. By contrast, based on a longer period (from 1979 to 2011), Bivens (2013) reveals that the United States has experienced rapidly growing wage and income inequality, and that growing trade with China alone is responsible for nearly one-sixth of the increase in income inequality.

A potential problem of Autor *et al.* (2013a) and many other researchers in this strand of literature is the control of technological changes in the United States. The increased wage gap and unemployment of unskilled workers in the United States could be instigated by technological changes and import competition from LWC. An identification problem thus emerges. Autor *et al.* (2013a) have endeavoured to include the 'percentage of employment in routine occupations' to alleviate this problem, but the problem is not fully resolved. They further address the problem in their subsequent two working papers. Based on the same county-level data, Autor *et al.* (2013b) reveal that competition from China tends to be geographically concentrated while the influences of technological changes are spreading among all regions. This observation can help solve the identification problem, thus allowing them to untangle the effect of technological changes to assess the influence of import competition from China (Autor *et al.*, 2013c). Their result indicates that import competition from China sharply increases manufacturing unemployment, particularly among workers without college education. This result confirms their earlier finding in Autor *et al.* (2013a).

The finding obtained by Autor *et al.* (2013a) is drawn based on differential effects at the county level. In contrast, Pierce and Schott (2012) conduct their analysis based on differential effects at the industry level. They consider the fact that the threat of reversion to higher import barriers was removed after the United States granted permanent normal trade relations (PNTR) to China. Using a difference-in-differences specification, Pierce and Schott demonstrate that industries with higher PNTR gaps experience larger employment declines, along with disproportionate increases in US imports from China, the number of US firms importing from China, the number of Chinese firms exporting to the United States and the number of US–China importer–exporter pairs. Their baseline regression indicates that the granting of PNTR reduces the employment growth of the average industry by 3.4% after 1 year, and by 15.6% after 6 years. This conclusion is further confirmed by the fact that corresponding effects are not significant in Europe because no policy shock is observed in the same period. The conclusion can also be confirmed from a different angle; that is, if Chinese exports to the United States reduce US employment, then the substitution effect will suggest that employment in China should increase accordingly. This view is verified by Ebenstein *et al.* (2011), who find industry-level evidence that Chinese employment growth is negatively related with that in the United States. In particular, employment growth in China has been largest in industries where US employment declines, which is consistent with the hypothesis of substitution between Chinese and US workers.

Interestingly, Autor *et al.* (2014) conclude that the employment effects are not only heterogeneous among regions and industries with differentiated exposures

to Chinese import competition, but also heterogeneous at the worker level. They report that import competition from China imposes heterogeneous labour adjustment costs for workers with different skill levels and conditions in the pre-competition shock period. Specifically, high-wage workers can adjust better, which results in lower earning losses. In addition, they are more capable of moving across employers and more likely to move out of manufacturing. By contrast, low-wage workers mostly stay in manufacturing and suffer increased earning losses. This differential effect on workers is consistent with the finding of Bivens (2013), who argues that low-skilled workers in the United States lose from trade, whereas the gains from trade are concentrated among college-educated, nonproduction and supervisory workers.²

Offshoring has become increasingly more important and more influential on the labour markets. Ebenstein *et al.* (2015) indicate that US offshoring to LWC is correlated with decreases in the wages of US workers, particularly those performing routine tasks and those forced to switch from high-wage manufacturing jobs to services. However, compared with the role played by trade, offshoring to China plays a quantitatively less important role in explaining wage declines in the United States.³

This strand of literature generally suggests that China's influence on the US labour market is very significant, as it affects different aspects of the labour market (i.e. employment, wage and labour participation) and various types of labour (skilled and unskilled).

2.1.2. Other developed countries

Although China's influence on the US labour market receives disproportionately more attention, some studies on other developed countries, mainly in Europe, exist.

In contrast to the case of the US labour market for which research is based on regional-level and industry-level data, many studies on European countries' experiences use plant-level data. Those studies generally demonstrate that competition from China has an overall negative effect on plant-level employment. In a study on 12 European countries, Bloom *et al.* (2016) reveal that increasing trade with China shifts employment from less technological advanced firms to more technological advanced firms.⁴ At the individual level, low-skilled workers are negatively affected the most as they are forced into unemployment or directly leaving the labour

² Trade's effect on the labour market has been dominated by the analysis of commodity trade and its effects on manufacturing workers. The paper by Liu and Treffer (2011) is an exception. Their empirical results reveal that the rising service imports from China and India from 1996 to 2007 had increased upwards occupational switching by 4% in the United States and downwards switching by 17%, boosted unemployment by 0.9%, and reduced occupational 'stayers' earnings by 2.3%.

³ Most, if not all, studies support the proposition that import competition from China contributes to the decline in employment in the United States. However, Scissors *et al.* (2012) argue that imports from China, particularly in apparel, toy and sorting industries, have supported over half a million jobs in the United States. These jobs are chiefly in the fields of construction, transportation, wholesale, retail and finance. They suggest looking at the job creation in service sectors that support increased imports.

⁴ Mion and Zhu (2013) conclude that offshoring to China by Belgian firms increases a small amount of non-production workers' employment in those firms.

force, whereas skilled workers, including college-educated, professional and technical workers, are modestly (and negatively) affected or not significantly affected. This pattern is also found by Utar (2014) for Denmark and Balsvik *et al.* (2015) for Norway. Utar (2014) finds similar unemployment pattern within Norwegian firms that less educated workers are unemployed while college educated workers retain. Balsvik *et al.* (2015) point out that the degree of the effect of competition from China on Norway's labour market is not as large as that on the United States. In particular, they contend that from 1996 to 2007, roughly 10% of the reduction in Norway's manufacturing employment is attributed to competition from China, which is merely half of what Autor *et al.* (2013a) find in the United States.

Like in the United States, China's effect on importing developed countries' wages is less clear-cut. Balsvik *et al.* (2015) find no significant evidence of the wage effect in Norway. Ashournia *et al.* (2014) analyse the wage effect of import competition from China using Danish firm-level data from 1997 to 2008. They reveal that competition from China reduces the domestic demand faced by firms, particularly the demand for low-skilled intensive products, and, thus, affects wages. Within job spells, their estimations demonstrate that roughly a 0.48% reduction in wage corresponds to each 1% increase in China's import penetration. By contrast, Utar (2014) indicates that importers and producers of multi-fiber-arrangement (MFA) goods in Denmark face increased competition from China following the removal of quotas, and, thus, experience an increase in the wage of skilled workers, including college-educated, professional and technical employees.

In contrast to most studies that report the significant effect of increased imports from China on local labour markets, Dauth *et al.* (2014) analyse the German case with industry-level data from 1988 to 2008 and contend that exposure to competition from China in those labour-intensive sectors exerts negligible effects on employment. A plausible explanation is that imports from China have replaced imports from other countries; hence, the China effect might cause job losses in the competing countries rather than in Germany. However, this case does not hold in the United States and many other countries in Europe.

In summary, the effects on the labour markets of other developed countries, chiefly European countries, are similar to but quantitatively smaller than those on the US labour market. The qualitative similarity is easy to understand as they are all developed countries, which face more or less the same type of competition from China. The difference in the degree of the effects could be a result of the labour market difference between the United States and Europe, as extensively documented in the labour economics literature.

2.2. Exporting deflation

The fact that China has massive production and supply of labour-intensive, low-technology and low-priced manufacturing goods to the world market constitutes a potential powerful deflationary force to the importing countries. This phenomenon is referred to as China exporting deflation. China's export effect on world prices is perhaps limited despite the fact that China is the largest exporter of

goods and commodities. Therefore, individual countries or prices of individual commodities have been emphasized. With regard to countries, most studies focus on developed countries, such as the United States, Japan and those in the EU. Regarding the types of products, researchers distinguish between the changes in import prices, producer prices and consumer prices.

Kamin *et al.* (2006) demonstrate that import competition from China has a modest effect on US import prices, reducing the overall US import price inflation by approximately 0.8% per year from 1993 to 2002. China's effects on 33 OECD nations are even smaller, lowering import price inflation by 0.10 to 0.25% on average per year. Broda and Weinstein (2010) point out some flaws in the traditional import price index (IPI) and propose a unified methodology to compute the IPI and consumer price index. The indices consider the introduction of new varieties of exports and quality adjustment. With their new measures and using HS nine-digit trade data, Broda and Weinstein report that the explosive growth of Chinese exports to Japan is accompanied by both the quality increase in existing export products and newly introduced export varieties. Thus, although imports from China have grown rapidly, they have not generated a significant effect on the aggregate prices in Japan. The new export varieties do have a significant effect on prices, but the effect is relatively small. The IPI adjusted for new varieties only increases 0.5% per year, which is slower than the unadjusted IPI.

With regard to producer prices, Kamin *et al.* (2006) find no significant effect on US producer prices. In contrast, using more refined data, including 325 manufacturing industries from 1997 to 2006, Auer and Fischer (2010) reveal that a 1% increase in the share of LWC in a sector of the US market will decrease the sector's producer price by 2.35%. Moreover, competition from China has a slightly stronger impact on the US producer prices than competition from other LWC. Similar effects are reported by Auer *et al.* (2013) using the data of European countries. They analyse the pressure of competition from China and other LWC (India, Malaysia, Mexico, Philippines and Thailand) on producer prices in European markets (Germany, France, Italy, Sweden and the UK). Their instrumental variable regression results indicate that a 1% increase in Chinese exporters' share of the market will reduce the producer prices by 3.8 to 6.8%, which is higher than the average effect of LWC (3.2 to 4.8%).

For consumer prices, Kamin *et al.* (2006) demonstrate that the effect of imports from China on the US consumer price is relatively small (roughly 0.1% or less). Using their new measures, Broda and Weinstein (2010) determine the effect on Japanese consumer prices to be insignificant. However, when exploring the composition of US imports from China, Broda and Romalis (2008) obtain different results. They match detailed US trade data with household consumption data between 1994 and 2005 to investigate the distributional effects of imports from China on the prices of non-durable consumption goods consumed by different income groups in the United States. Chinese imports chiefly concentrate on low-quality, low-price non-durable consumption goods that are heavily consumed by poor people in the United States; hence, imports from China are more likely to help reduce the average price of consumption of the poor. Their

empirical findings confirm that imports from China reduce the relative price index of the poor by approximately 0.3% per year, accounting for one-third of the relative price drop of the poor. This effect can also offset roughly 30% of the increased income inequality in the United States.

Bugamelli *et al.* (2010) focus on a different aspect of price changes. They use Italy's manufacturing firms' data over 1990 to 2006 to examine the pro-competitive effect of imports from China on Italy's price dynamics. Their results indicate that a 10% increase in the Chinese *share* in imports will decrease the price dynamics by 0.30 to 0.35%, whereas the same increase in overall imports will only decrease the price dynamics by 0.06%. The price change is larger in less technological sectors and smaller firms.

In summary, the general price reduction effect from China's competition is low. Thus, the claim that China exports deflation is supported. Nonetheless, some evidence shows that China's export growth significantly lowers the prices of certain products in several importing countries. Potential measurement errors are included in the calculated prices, and robust and accurate results are difficult to obtain.

2.3. *Resource reallocation and productivity*

The idea that international trade induces resource reallocation is old. However, the understanding about reallocations varies at different stages of the literature. Traditional trade theories deal with resource reallocation across industries (Ricardian theory and Heckscher–Ohlin theory). The most recent trade theories focus on resource reallocation across firms within the same industry (Melitz, 2003; Bernard *et al.*, 2006) and even across products within the same firm (Bernard *et al.*, 2010, 2011). These trade-induced resource reallocations have strong implications for productivity and technology progress. In this subsection, we review the set of empirical studies related to China's trade and its effects on resource reallocation.

Bernard *et al.* (2006) investigate trade-induced resource reallocation in the United States from 1977 to 1997. They document the fact that when facing high exposure to competition from LWC, including China, plants are more likely to switch to industries with lower exposure and, thus, will shift resources from most labour-intensive to capital-intensive industries. Given the trade-induced resource reallocation, Ebenstein *et al.* (2011) point out that the role of trade from China in explaining the productivity increase of the United States has been underestimated. The increase in Chinese industry employment is determined to be highly correlated with unit labour cost decline and productivity growth in the United States, indicating the role of China's trade in the productivity growth of the United States. The increase in routine occupations and the decrease in the manager-to-worker ratio in China are all negatively correlated with those in the United States. These results are obtained from industry-level data.⁵

⁵ This negative correlation could be an outcome of increased bilateral trade or increased offshoring activities.

Competition from China has heterogeneous effects on firms' likelihood to survive, grow and achieve technological progress, which, in turn, induces substantial resource reallocation across firms within industries and significantly affects skill upgrading and technological progress across firms. Bernard *et al.* (2006) conclude that plant growth and survival rates suffer from the competition of LWC, including China. In industries with high exposure to competition, capital-intensive or skill-intensive plants have relatively higher plant growth and survival rates than low skill-intensive plants. Similarly, Bloom *et al.* (2016) provide evidence that Chinese import competition decreases the surviving probability and employment in low-technology firms, whereas high-technology firms are somewhat 'shielded' from the competition effect. Martin and Mejean (2014) suggest that cross-firm resource reallocation also occurs in third markets. They demonstrate that demand for French exports in third markets is shifted from low-quality French exporting firms to high-quality firms. Consequently, the average quality of French exports increased by 10 to 15% from 1995 to 2005. The quality of French exports improves the most in markets where quotas on Chinese imports had been removed after 2002, and thus, where competition from China has increased the most. Mion and Zhu (2013) use Belgian firm-level data from 1996 to 2007 to investigate firm responses to competition from different countries. They reveal that firm survival rate is not significantly affected by import competition from China, firm employment is negatively affected, but skill upgrading in low-technology manufacturing industries is positively affected.

Recent studies also stress within-firm resource reallocation. Bernard *et al.* (2006) indicate that firms will adjust their product mix in response to import competition. Utar (2014) confirms this adjustment and reveals that firms change their product mix by focusing away from products in which China has higher comparative advantage. Bloom *et al.* (2016) contend that firms will innovate more when facing competition from China. They use 12 European countries' firm-level data matched with patent data and take the quota removal by China's accession to the WTO as an instrumental variable in their empirical analysis. Their results indicate that competition from China can explain approximately 15% of European technology upgrading over 2000 to 2007. They further explain that this upgrading occurs through two channels, namely, within firms and across firms, which together account for roughly half of the total effect. Chinese import competition increases absolute innovation (more patents, higher IT intensity and higher TFP) within surviving firms, particularly in their output markets. However, imports from developed countries do not have a significant effect on innovation.

Competition from China will also compel firms in low-technology manufacturing industries to hire more non-production and highly educated workers. In this regard, Mion and Zhu (2013) conclude that import competition from China accounts for 27 and 48% of the total observed increase in the share of non-production workers and that of highly educated workers, respectively, in Belgium's low-technology manufacturing sector. Moreover, competition might drive firms to adopt different organizational forms. Utar (2014) asserts that

the skill intensity of base-level occupations in Danish firms has increased, indicating a flattening firm organization. Competition from China also forces firms to adapt to more flexible and skill-oriented organizational forms because of the distributional effects.

All studies indicate that China's competition results in resource reallocations between industries and between firms in the importing countries. The latter type of resource reallocation (i.e. firm-level responses) is consistent with the general finding in the recent trade literature. For example, Bloom *et al.* (2016) point out that within incumbent firms, productivity growth typically accounts for at least as much as these between-firm reallocation effects. Trade will induce firms to hire more skilled and more educated workers, to innovate more, and to employ production processes with higher technology and productivity, as well as to adopt flatter organizational forms to increase productivity and quality.

3. ASIAN COUNTRIES/ECONOMIES

China's effects on its neighbouring Asian economies could be very different from those on the others for several reasons. First, Asian economies have geographical proximity to China, and some even share similar cultural, language or common borders with China. Second, in recent years, China and other Asian economies have increasingly integrated with each other into the regional and global production networks.⁶ Third, China has comparative advantages in low-technology and labour-intensive products, but it has increased exports of middle-technology and high-technology products. Fourth, China has been the 'world processing factory', which implies a huge demand for imports of intermediate inputs and massive exports of final products. Fifth, different Asian economies are at different stages of economic development and, thus, compete with China at various levels.

Studies principally focus on the following two questions. First, what are the overall effects of China on exports and imports of other Asian economies, at both bilateral and regional levels? Second, what are the differential effects of China on other Asian economies at different development levels, such as India, newly industrialized economies (NIEs)⁷ and Association of Southeast Asian Nations (ASEAN).⁸ Researchers have adopted different approaches and used various data sets to conduct their analyses. The basic methods can be classified into the following three groups:

1. *Measure.* Research using this approach attempts to measure the different features of trade in comparing China and other Asian economies with regard to their competitiveness and complementarity. The most direct and simple measure is 'relative market share dynamics'. Another important measure is

⁶ He and Liao (2012) find that the Asian countries sustain a relatively strong independent business cycle among themselves. Outputs of Asian countries are less synchronized with global factors than those industrialized economies.

⁷ NIEs include Korea, Singapore, Hong Kong and Taiwan.

⁸ ASEAN includes Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, the Philippines, Singapore, Thailand and Vietnam (<http://www.asean.org>).

‘export structure similarity’. Revealed comparative advantage (RCA) is also commonly used. These measures provide only rough evidence and certain types of correlations, but not causality. Moreover, quantitative importance cannot be assessed using this approach.

2. *Computational General Equilibrium (CGE) Model*. Most CGE models in this literature are derived from the Global Trade Analysis Project (GTAP). To provide quantitative outcomes from a general equilibrium perspective, researchers conduct simulations based on different versions of GTAP with various data sets and slightly modified assumptions. This approach can be used in performing counterfactual analysis and in forecasting. However, strong assumptions are often required. Different methods for calibrating the models also matter for the results.
3. *Regression*. Regressions, particularly those based on gravity models, constitute the main body of this line of literature. Using more detailed data and with instrumental variables, this approach works better in establishing the causality of China’s effects on other Asian economies.

In this section, we discuss the literature on China’s effects on Asian economies according to the aforementioned three different approaches.

3.1. *Measures approach*

Using a variety of measures, researchers reveal that the potential threat of China is not as large as its surge in exports might suggest (Fernald *et al.*, 2003; Lall and Albaladejo, 2004; Shafaeddin, 2004; Li and Song, 2005). Two major explanations have been proposed.

The first explanation emphasizes the importance of increasing demand by China for exports of other Asian economies. Lall and Albaladejo (2004) report that China has provided exporting opportunities for Asian economies in the 1990s as its import growth outpaces export growth. Shafaeddin (2004) demonstrates that NIEs face severe competition from China, but also benefit from increased demand for parts and components from China, which is sufficiently large to offset China’s negative effects in the short and medium run. Li and Song (2005) conclude that although China exerts great pressure on other East Asia economies, it also provides huge exporting opportunities for them. More industrialized economies, such as Japan, Korea and Taiwan, benefit more from increased import demand from China than less developed ASEAN economies do.

The second explanation states that competition from China is primarily concentrated on low-technology or labour-intensive sectors in Japan, NIEs and ASEAN (Fernald *et al.*, 2003; Schott, 2004; Shafaeddin, 2004; Li and Song, 2005; Qureshi and Wan, 2008). However, such competition is partially the outcome of the industrial development strategies of more advanced economies, which intentionally shift the process or production of those industries to China while keeping other core processes and industries in their own economies. Thus, competition does not have large negative effects on the welfare of those economies.

Although the overall negative effects are small according to this set of studies, the distributional effects substantially vary across products, industries and countries within the region.

At the country level, Qureshi and Wan (2008) calculate the coefficient of competition and coefficient of conformity at Standard International Trade Classification (SITC) three-digit and four-digit level for exports to capture the similarity in the export structure of two countries, which are used to represent the potential competition. They also calculate the trade complementarity index to capture the potential complementarity between two countries, which can be used in assessing the potential for expanding trade between these countries. Their measures based on data from 1995 to 2005 suggest that with regard to bilateral trade, competition between China and Japan, Korea, Hong Kong, Malaysia, India, Indonesia, Thailand and Vietnam is relatively severe. However, China has high trade complementarity with Korea, Singapore, Malaysia, Thailand and India. In third markets, China is chiefly competing with India, Hong Kong, Korea, Malaysia and Thailand. Using market share as a measure, Fernald *et al.* (2003) conclude that the 'New Tigers' (i.e. Indonesia, Malaysia, Philippines and Thailand) experience low market share gains. Shafaeddin (2004) argues that South Asian economies face competition from China in labour-intensive manufacturing goods but receive no increased demand from China for their exports.

At the product or industry level, competition from China principally focuses on low-technology products. The potential threat in middle-technology products is increasing, but China is complementary to other Asian economies in high-technology products (Fernald *et al.*, 2003). However, in third markets, Shafaeddin (2004) indicates that NIEs and ASEAN also face competition from China in capital goods, and Qureshi and Wan (2008) suggest that competition is chiefly concentrated in particular industries, such as clothing, electronic, telecommunication and electronic equipment, and statistical and office machinery. Humphrey and Schmitz (2006) investigate more detailed cross-industry distributional effects. They conclude that Central Asia benefits from increased demand by China for energy, but energy importing economies are hurt because of the increase in price. For final manufacturing goods, no clear evidence shows that China has displaced other Asian economies because the exports of most of these economies have also increased since the 1990s. The exceptions are Japan, Korea and Taiwan, but they should not blame competition from China because their export decline is a consequence of their industry development strategies mentioned previously. Humphrey and Schmitz (2006) also express concern that with the development of China, with the transition from assembly to more high-technology production, the complementarity between China and the more advanced economies in Asia might be replaced by competition in the future.

Gaulier *et al.* (2007) point out that the vertical integration among Asian economies is important for understanding the role of China. The export growth of China from 1993 to 2003 is heavily related to its import growth and has increased intra-Asia trade. However, the effect of China's export growth is small, and Asia continues to depend on western markets for final goods trade.

3.2. *Computational general equilibrium approach*

The major objective of using the CGE approach is to quantify the effect of the rise of China on other Asian economies. Researchers use different versions of the GTAP model in performing their analyses.

Several studies conclude that the welfare of most Asian economies, particularly the more advanced ones, slightly benefits from China's integration into the world market, whereas some developing countries suffer different degrees of losses. In Ianchovichina and Walmsley (2005) simulations, Japan and most NIEs, except Singapore, benefit from China's entry to the WTO. The welfare gains range from US\$86m for Hong Kong to US\$1535m for Taiwan. These economies' rates of returns to capital and terms of trade improve because of the increased exports of high-technology and capital-intensive products to China and increased imports of cheap labour-intensive products from China. The improvement in returns to capital ranges from 0.08 to 0.33%, and that in terms of trade from 0.03 to 0.38%. In the IMF (2004) simulations, the predicted cumulative improvement in the welfare of NIEs by 2020 is 0.2% on average, and improvement in terms of trade is 0.5% if China's productivity, employment and physical capital are assumed to increase more compared with the case in which China's output share in the world remains fixed at the 2002 level. Both the exports and imports of NIEs benefit from the rapid growth of China, increasing by 2.2 and 2.9%, respectively. By contrast, South Asia faces severe competition from China. The welfare of those countries decreases by 0.2% and output decreases by 0.5%. These countries do not significantly benefit from the increased demand for their exports by China or the increased supply of cheap labour-intensive products from China because their exports only increase by 0.1%, their imports decrease by 2.3%, and terms of trade deteriorate by 1.0%.

At the industry level, competition from China commonly occurs in labour-intensive industries, such as textiles, clothing and apparel (IMF, 2004; Ghosh and Rao, 2010), and competition in these industries is pervasive across all Asian economies regardless of development stage. Developing economies in Asia simultaneously face competition from China in some capital-intensive industries, such as automobiles (Ianchovichina and Walmsley, 2005). In particular, according to the IMF (2004) simulations, the effects of China's rapid growth on labour-intensive manufacturing outputs are -3.2% for NIEs, -6.6% for ASEAN and -0.9% for other South Asian countries. For textiles and clothing outputs, the effects are -11.9% for NIEs, -12.0% for ASEAN and -8.0% for other South Asian countries. In a more refined GTAP model simulation, Ianchovichina and Walmsley (2005) reveal that in the textile industry, Japan and NIEs, except Singapore, benefit, with the benefit ranging from US\$339m for Hong Kong to US\$5090m for Taiwan. In the apparel industry, Japan benefits by approximately US\$865m, but NIEs suffer losses, ranging from US\$89m for Singapore to US\$2940m for Hong Kong. Developing countries in South-East Asia, including Indonesia, Vietnam, Malaysia, Philippines, and Thailand, all suffer losses in textile output (US\$61m to US\$970m) and apparel output (US\$768m to US\$1337m). However, according to calibration and simulations in Ghosh and

Rao (2010), in which the removal of trade barriers in textile and apparel industries has been considered along with China's entry to the WTO, Asian countries have large gains from textiles and apparel. They also single out India in their study and reveal that in textiles, India gains 19.0, 9.9 and 7.5% in exports, imports and value added, respectively, whereas the rest of Asia gains 41.5, 21.9 and 19.3%, respectively. For apparel, India gains 37.3, 8.2 and 31.5% in exports, imports and value added, respectively, whereas the rest of Asia gains 59.7, 10.0 and 18.3%, respectively. These high gains are achieved partially by the promotion of bilateral trade within Asia.

Asian economies also face competition from China in some skill-intensive and capital-intensive industries, such as automobiles. According to the IMF (2004), the skill-intensive manufacturing outputs of Asian economies are slightly and negatively affected by the rapid economic growth of China. Specifically, the effects are -0.7% for the NIEs, -0.6% for ASEAN countries and -0.1% for other South Asian countries. Ianchovichina and Walmsley (2005) report that Japan and NIEs suffer losses in their automobile outputs because of China's entry to the WTO. China's entry has also decreased the automobile output of Singapore down to US\$223m and that of Japan to US\$6130m. Other East Asian developing countries, such as Indonesia, Vietnam, Malaysia, the Philippines and Thailand, also suffer small losses in their automobile outputs, ranging from US\$2m to US\$178m.

3.3. Regression approach

Regression analysis requires more data and more rigorous techniques. In some cases, the identified problems must be addressed as soon as possible.

Using data from 1981 to 2001, Fernald *et al.* (2003) conclude that China's exports are positively correlated with those of other emerging Asian economies and that little evidence shows that China's exports are 'crowding out' those of other emerging economies. During this period, the exports of China and other emerging Asian economies all heavily rely on the demand conditions of their major trading partners, such as the United States, the EU and Japan. However, by employing gravity models, Eichengreen *et al.* (2007) find that Chinese exports have an insignificant effect on those of other Asian countries. To further understand the role of China's growth, Eichengreen *et al.* distinguish the effect of the increased imports demand in China on the country's export surge. They also disaggregate commodities into consumer, capital and intermediate goods. Their regression results indicate that China's exports (chiefly consumption goods) tend to crowd out those of other Asian economies, particularly the third countries, between 1990 and 2003. Based on these findings, they suggest that those developing countries that mainly export consumption goods are more severely and negatively affected by China's exports compared with the more developed Asian economies that primarily export capital and/or intermediate goods. However, using a gravity model with data from 1990 to 2003, Greenaway *et al.* (2008) observe a crowding-out effect of China on the average imports of Asian economies,

which contradicts the findings of Eichengreen *et al.* (2007). These contrasting conclusions may be attributed to the different methodologies that have been adopted in these two studies. For instance, in contrast to Eichengreen *et al.* (2007), Greenaway *et al.* (2008) exclude year dummies in their pooled estimation. Greenaway *et al.* (2008) also divide Asian economies into low-income, middle-income and high-income groups and run their regressions with each group sample. They reveal that China's exports significantly affect those of high-income Asian economies (e.g. Japan, Korea, and Singapore), but insignificantly affect those of low-income and middle-income Asian economies. The crowding-out effects are significantly higher when Hong Kong's exports are considered, thereby confirming the important role of Hong Kong as a re-exporter for China. This conclusion contradicts the inference of Eichengreen *et al.* (2007) that less advanced countries are more severely affected by China because their exports are principally concentrated on consumption goods. However, when the year dummies problem is put aside, the findings of Greenaway *et al.* (2008) may be preferable because they provide direct evidence rather than inference. Moreover, their results can be explained by the observation that China's competition with more advanced economies is largely caused by the shift of relinquishing industries from those economies to China. This phenomenon has also been observed by other studies that have used the two abovementioned approaches. For instance, by analysing the export growth rate in third markets, Fernald *et al.* (2003) demonstrate that although the increased imports demand in China for capital and intermediate goods has benefited the more advanced Asian economies, China also crowds out these economies, especially NIEs, in several industries, such as footwear, apparel and household products.

Most studies view China as a competitor when examining its effects on other countries. These studies chiefly focus on horizontal specializations in trade in which countries produce and trade final goods yet ignore the increasingly important vertical integration of China into global production networks, which is complementary to the production processes of various countries. By contrast, Athukorala (2009) treats China as part of the global or Asian production value chain to fill the research gap. Athukorala uses disaggregate SITC five-digit trade data covering the years 1995 to 2005 and then divides these data into parts, components and final goods. The regression results indicate that the crowding-out effect of China is vastly exaggerated and that no evidence can prove that the competition from China induces proportionate losses for other Asian developing countries. In fact, China's vertical integration into the production networks as an assembly centre is complementary to other East Asian economies' production of parts and components. China's expansion in the export of labour-intensive manufacturing goods is principally limited to NIEs. The rise of these industries in China is largely at the expense of NIEs that strategically shift some low-skill or low-value-added labour-intensive industries into China to focus more on capital-intensive and high value-added industries.

Using a structural VAR model, Abeyasinghe and Ding (2003) reveal that China has a significant positive effect on the GDP growth of its neighbouring economies and that such an effect is already significant before China has entered

the WTO. China's entry into the organization has only reinforced and magnified such a positive effect.

In contrast to the other studies, Coleman (2007) focuses on China's effect on the other countries' prices. The author builds a model to explain such an effect and empirically examines how the emergence of China affects the neighbouring economies' prices and other fundamentals. The sample covers 1970 to 2003 data from Japan, Korea, Singapore, Taiwan, Hong Kong, India, Malaysia, Indonesia, Thailand, Philippines and Australia. Coleman reports that the price increases that are induced by China's exports tend to fall. The surrounding economies and regions, regardless of their development stages, generally tend to shift their employment away from manufacturing goods industries because of the large increase in the world export share of China for production services. These empirical results also confirm the model's prediction that the elasticity of a country's real GDP on the world prices for Chinese exports is monotonically related to China's wage share in the production of these goods.

Given that different approaches have been used to investigate the effects of China on other Asian economies, various conclusions on the aspects of such effects have been generated. The aggregate effect of China on the welfare of other Asian economies is positive yet quantitatively limited (i.e. the effect of China on output is either modestly positive or negative). Intraregional trade is mainly augmented because of the increased trade of intermediate goods that is induced by China's integration into the Asian and global production network. Overall, the Asian trade, particularly in final goods, is still driven by the demand from the US and Europe.

At the individual country level, more advanced economies, such as Japan and NIEs, benefit from China's increased import demand, thereby offsetting some negative effects of the country. South Asian economies are negatively affected by China's growth and competition in third markets. Despite facing strong challenges from China in the textile and clothing sectors, the other Asian economies are both positively and negatively affected by China's economic growth.

4. OTHER DEVELOPING COUNTRIES

China's integration into the world may have various effects on each developing country. On the one hand, the increasing demand of China for raw materials and intermediate inputs can foster the production and exports of developing countries that have comparative advantages in producing these goods. The exports of cheap manufacturing goods by China can also benefit consumers from developing countries. On the other hand, the comparative advantages and exporting structures of China are overlapping with those of many developing countries. Therefore, the explosive growth in Chinese trade also poses a challenge for developing countries. For example, the increasing imports of cheap manufacturing goods from China have threatened the domestic production of some developing countries and have 'squeezed' out their exports in third markets. These concerns are particularly acute for countries that have devoted most of their resources to developing textile and cloth export structures under the

protection of the MFA quota system. Given that this system has now been removed, these countries are faced with increased direct competition from China (Jenkins, 2008b). Along with the increased demand for raw materials from China, such a competition may also induce the 'deindustrialization' or 'primarization' of the production and exports of developing countries and then hinder these countries from upgrading the quality of their manufacturing goods (Jenkins and Barbosa, 2012). Each developing country will be affected differently depending on its economic positions in the world market relative to China.

Hanson and Robertson (2008) further investigate this issue using a gravity model to decompose trade into three components that capture the exporting countries' supply conditions, importing countries' demand conditions and bilateral trade costs. Using these measures, they examine how China's export growth affects 10 developing countries that specialize in manufacturing goods.⁹ China's export growth only has a modest negative effect on the exports of these countries between 1995 and 2006. Therefore, even if China's export supply conditions remain unchanged from 1995 to 2005, the demand for exports from these 10 countries will only increase between 0.8 and 1.6%.

However, Jenkins (2008b) argues that most traditional indices only measure the competition between China and other developing countries according to the similarity of their export structures. Given that Chinese exports are more diversified and significantly larger than those of other developing countries, these indices will generate biases, thereby underestimating the competition effect from China. Therefore, in addition to traditional measures such as RCA, coefficient of conformity and index of trade competition, Jenkins proposes two new indices for solving the measurement problem. The basic idea of Jenkins is to calculate a country's export share of those products that are primarily exported by China. The RCA and growth rate of exports are used to identify the products in which China has global competitiveness. Products with $RCA > 1$ are considered statically competitive, whereas those products whose export growth exceeds the world trade growth are considered dynamic or potential global 'threats'. These measures, which are calculated using the export data of China and of 18 developing countries in Asia, Africa and Latin America, show that the competition from China between 1996 and 2002 tends to be higher than what had been indicated by previous researches.

China's effects are not homogenous. Thus, we investigate the effects of China on different groups of countries and regions. The first of these groups consists of the Latin American and Caribbean (LAC) countries, among which special attention will be paid to Mexico and Brazil. The second group comprises the African countries.

4.1. *Latin American and Caribbean countries*

Previous studies conclude that the trade structures between China and most LAC countries, except Mexico, have remained dissimilar until 2005. The

⁹ These 10 developing countries are Hungary, Malaysia, Mexico, Pakistan, Philippines, Poland, Romania, Sri Lanka, Thailand and Turkey.

major export products of LAC countries do not overlap with those of China; thus, the majority of the LAC countries (including Argentina, Brazil, Chile, Colombia, Peru and Venezuela) are not severely affected by China. Lall *et al.* (2005) use trade data from 1990 to 2002 and reveal that the trade structures of LAC countries are largely complementary instead of competitive with that of China. Therefore, some LAC countries have benefited from trading with China, particularly in primary and resource-based products, such as soybeans, soy oil, wood, iron ore and steel. This finding is also confirmed by Qureshi and Wan (2008). However, these benefits are quantitatively small (Lall *et al.*, 2005).

Interestingly, recent studies have mostly suggested that China poses a threat more than an opportunity to LAC countries.

Jenkins (2008a) argues that the traditional measures for competitiveness are inadequate to capture the competition from China. Jenkins proposes a new index that explicitly combines the extent of the threat and intensity of such competition to measure the competitive threat from China. Applying this new index to data from 1996 to 2006, Jenkins indicates that Latin America's exports to the US market have been losing their market share to Chinese exports since 2001, particularly in manufacturing goods. This trend of the loss seems to increase over time. Freund and Ozden (2009) also demonstrate that Chinese export growth has a negative effect on the LAC countries' exports to third countries. For an average product with a Chinese market share of 10% and export growth of 20%, the LAC export growth will be reduced by 0.6%. Some evidence shows that larger imports from China can also decrease the average exports prices of LAC countries. In particular, a 10% decline in the export prices from China results in a 0.4% price decline in the export prices from LAC countries. Hanson and Robertson (2009) reveal that the increased export–supply capacity of China and the decreased import–demand condition in the United States have contributed to a slowdown in the growth of the manufacturing exports of four major Latin American countries (i.e. Brazil, Argentina, Mexico and Chile). If the export–supply capacity of China has remained unchanged since 1995, these four countries' exports will be 0.5 to 1.2% higher between 1995 and 2000 and 1.1 to 3.1% higher between 2000 and 2004. If the US growth from 2000 to 2004 has remained the same as that from 1995 to 2000, the exports of the four LA countries will be 0.2 to 1.4% higher. Using evidence from Chilean plants, the regression results of Álvarez and Claro (2009) indicate that the import competition from China has increased the probability of exit of Chilean plants and has decreased their employment growth.

Some researchers also argue that the competition from China may force LAC countries to upgrade their product quality and productivity, thus allowing them to climb up the 'quality ladder'. However, empirical studies find no significant positive effect of competition on product quality. Freund and Ozden (2009) assert that although the competition from China has improved the quality of several industries such as textiles, Chinese export competes primarily in high-wage LAC products and can potentially hamper

LAC countries from moving up the quality ladder. Although Chilean plants are facing direct competition from China, no significant evidence of 'trade-induced' output upgrading has been found.

López-Córdova *et al.* (2008) employ a different approach for estimating the elasticity of substitution between Chinese and Latin American exports to the United States using Harmonized System (HS) six-digit trade data to assess the competition between these exports under different proposed policies. The appreciation of the Chinese currency, renminbi, would reduce the Chinese exports to the United States, but will increase the exports from Latin America. Establishing a free trade area in the Americas will help Latin America and other related countries to compete with China by increasing their exports and decreasing China's exports to the US market. Removing MFA quotas for China may sharply increase the exports of China to the United States (40%), but decrease those of Latin America and other regions. Therefore, China is facing a significant competition from Latin America in the US market and the country's gain of market share in the US market primarily comes at the expense of Latin America and other countries.

Even within LAC countries, different countries face various forms of competition from China. Special attention must be paid to Mexico and Brazil, which are the largest of all LAC countries and are highly similar to China in terms of development stages, comparative advantages, production structures and export patterns.

4.1.1. Mexico

Researchers have not reached a consensus on whether Mexico is facing severe competition from China. Instead, they have investigated how different industries and firms in Mexico are affected by such competition.

At the industry level, the Mexican exports in the US market have been losing their dynamism since China's entry to the WTO. The relative market share of Mexican exports in the United States is either growing less rapidly than that of Chinese exports or declining (Gallagher *et al.*, 2008). Utar and Ruiz (2013) investigate the distributional effects across various Mexican industries and reveal that Mexican maquiladoras in all industries are negatively affected by the competition from China, with the plants in unskilled labour-intensive industries suffering the most. The competition also induces resource reallocation across several industries, which promotes industrial upgrading.

At the firm level, Iacovone *et al.* (2013) conclude that the substantial resource reallocation across Mexican firms is driven by competition from China. In particular, when faced with import competition from China, smaller firms in Mexico decrease further in size and tend to quit, whereas larger firms are not significantly affected. Smaller Mexican exporters also suffer from a similar level of competition in the US market. Larger firms benefit from the expanded access to cheap Chinese imported intermediates. Generally, the import competition from China benefits the larger (and supposedly more productive) firms in Mexico, but negatively affects the smaller firms. These

effects vary not only across each firm, but also across each plant within firms. Utar and Ruiz (2013) focus on the reaction of Mexican maquiladoras to the Chinese competition from 1990 to 2006. Both employment and plant growth are negatively affected by Chinese imports at both the intensive and extensive margins. Interestingly, although significant reactions have been observed within firms, Utar and Ruiz fail to find any significant difference in the reactions of firms within industries.

At the product level, Iacovone *et al.* (2013) argue that when facing import competition from China, the imports of marginal products will shrink and more easily quit, whereas the imports of core products are relatively impervious. Core products benefit from the import of intermediate goods from China, which indicates the existence of market share reallocation toward core products within firms.

4.1.2. Brazil

Although China has a limited effect on Brazil's domestic market, considerable competition has been observed in third markets. Jenkins (2014) demonstrates that from 2000 to 2010, Brazil's exports to most destinations are negatively affected by Chinese exports. Brazil has faced large losses in the markets of Chile, the United States and Venezuela, but has faced small losses in Mercosur countries and the EU. Brazil's loss in Latin American markets is also confirmed by Gomes de Lima *et al.* (2012), who indicate that China's share of imports and trade intensity index in Latin America increased between 1994 and 2009, especially from 2007 onwards, whereas that of Brazil has stagnated between 1994 and 2008. This finding indicates the displacement effect of China on Brazil in Latin American markets. Jenkins and Barbosa (2012) derive the same conclusion, but argue that the absolute effect of China on Brazil is small considering the latter's small scale of exports. Therefore, competition is widely observed in third markets, but researchers have presented contrasting findings on the extent of the effects of such competition.

With regard to the distributional effects at the industry or product level, some researchers reveal that a wide range of industries, not only the labour-intensive ones, are affected by the competition from China. Jenkins and Barbosa (2012) point out that in Brazil's domestic market, China's exports have a high penetration in electronic industries, but have a below average penetration in traditional industries. In third markets, Brazil loses its market share to China in both high-technology and traditional industries. Gomes de Lima *et al.* (2012) argue that in Latin American markets, Brazilian exports show advantages in natural resource-intensive sectors, tractors, agricultural equipment and aviation, whereas Chinese exports exhibit advantages in all other industries. Jenkins (2014) also demonstrates that a wide range of products and industries are negatively affected by Chinese exports. Low-technology products suffer the greatest losses, whereas some high-technology products, such as colour television sets, mobile phones, and digital processing equipment, in the United States and several Latin American countries also suffer losses.

4.2. *Africa*

Jenkins and Edwards (2006) investigate the effect of the rise of China and India on 21 Sub-Sahara Africa (SSA) countries and conclude that the export structures of SSA countries are significantly different from those of China and India. Therefore, the threat of competition that has been observed in Asian economies and LAC countries is not a serious concern in Africa. They also determine that the effects from Asia in general and from China in particular are heterogeneous among SSA countries. For example, Angola, Nigeria and Sudan are important exporters to Asia, Ghana, Ethiopia, Kenya, Tanzania and Uganda are the main importers of Asia, and Lesotho faces competition from China in third markets.

By contrast, He (2013) argues that compared with the imports from developed countries, such as the United States and France, SSA countries benefit more from the imports of Chinese goods. Using UNIDO panel data of 28 industries and 17 SSA countries between 1976 and 2004, He (2013) indicates that Chinese imports positively affect the exports of all SSA manufacturing sectors, and such an effect is stronger than that of imports from France and the United States. He also demonstrates that when the importing country has a limited absorptive capability and/or the imported intermediate goods are highly substitutable to local production, the country will benefit more from importing goods from a developing country that has better technology (i.e. China) than from a developed country that has very superior technology (i.e. United States). Therefore, Africa can substantially benefit from increasing its trade with China.

Some studies also indicate that Chinese exports have a significant competition effect within Africa and in third markets. Giovannetti and Sanfilippo (2009) contend that Chinese exports compete with African exports in their major markets, including the United States, the EU and other African countries. An increase in the share of China in these markets is associated with a decrease in the share of Africa. Chinese exports to SSA have also displaced the intraregional trade in Africa, particularly intra-SSA trade. At the industry level, textiles and clothing sectors as well as high-technology industries, such as machinery and equipment, chemicals and non-metallic mineral products, are suffering from the displacement effects of their competition with China. Kaplinsky and Morris (2008) provide similar industry-level evidence by showing that the textile and clothing industries of SSA countries are facing considerable competition from Asian producers, particularly China, in both domestic and global markets.

Compared with other SSA countries, South Africa has been significantly affected by the competition from China. Jenkins and Edwards (2015) reveal that South Africa's exports in all major markets have been affected by Chinese exports. For instance, South Africa's exports to other SSA countries, the EU and the United States have all decreased by 10, 7.8 and 5.6%, respectively, in 2010. According to Edwards and Jenkins (2014) calculation that focuses on SSA markets, the exports of South Africa's existing and new products have been negatively affected by the competition from China. South Africa's exports to other SSA countries in 2010 would be 20% lower if the country was affected by China at the same degree as that suffered by other SSA countries. Such a

negative effect has been principally attributed to the intensive margin, whereas the extensive margin is quantitatively less important. Although 'crowding out' effects have been observed in these third markets, such effects are relative because the absolute exports of South Africa to these markets have all increased over the years. Edwards and Jenkins (2015) demonstrate that the imports from China have significantly displaced not only the imports from other countries, but also the domestic production of these countries.

At the product level, almost all manufacturing exports of South Africa are negatively affected by the competition from China, with the low-technology and middle-technology products suffering the most (Jenkins, 2014; Jenkins and Edwards, 2015). Edwards and Jenkins (2015) point out that although the imports from China have significantly displaced the domestic production and exports of South Africa, the country's exports to China have no significant effects on the latter's industrial growth. These findings indicate an overall negative effect of Chinese exports on the manufacturing employment in South Africa. The imports of Chinese products also lower the producer prices and moderately increase the consumer prices in South Africa.

5. GLOBAL INFLUENCE

This section reviews the studies that investigate the larger picture of China's influence. We initially examine the overall effect of China, then the distributional effects across regions, and finally the distributional effects across sectors.

5.1. Overall effects

Researchers have agreed that China's growth and integration into the global economy can benefit the entire world, but the degree of its influence may be limited. For example, IMF (2004) simulations show that China's rapid growth in output and trade has a positive effect on global welfare, but such an effect is relatively small across different regions (from 0.1 to 0.7%).¹⁰ This finding has been confirmed by Ghosh and Rao (2010), who have demonstrated that the overall effect of China's entry to the WTO, together with the removal of MFA quotas, on other countries' real GDP is positive yet quantitatively small. For example, the real GDP of Canada, the United States and the EU will only increase by 0.06, 0.02 and 0.01%, respectively. Based on a GTAP model, Ianchovichina and Martin (2004) demonstrate that the global gains of China's accession into the WTO total US\$74 166m in absolute value.

From a different perspective, Dimaranan *et al.* (2007) suggest that the increased variety and improved quality of Chinese and Indian exports can substantially promote world welfare. Levchenko and Zhang (2012) analyse how the trade integration of emerging giants, namely, China, India, and Central and Eastern Europe, affects the aggregate and sectorial welfare gains of the

¹⁰ South Asia is excluded. This region is the only one to suffer losses from China's growth.

world between 2005 and 2007. They reveal that the aggregate welfare gains to the rest of the world are 0.37%. Bloom *et al.* (2014) assess such an effect from a different perspective by incorporating the dynamic effect of LWC import competition on firm innovation in a product cycle model. In this model, trade liberalization will promote the long-term growth of Northern firms because of the increased profits from sales in the South. However, in the short run, production factors can only be reallocated within firms, thus reducing the opportunity cost of innovation and generating an additional ‘trapped factor’ effect. The calibration results of Bloom *et al.*, who focus on OECD growth between 1997 and 2006, indicate that trade integration with LWC has increased the OECD’s long-run growth rate from 2.0 to 2.4%, which is higher than the conventional calibrations of static trade models that ignore the dynamic effect of trade on innovation. Approximately 0.2% or half of this increase can be attributed to China alone. The additional short-run effect that is incited by the ‘trapped factor’ increases the short-run growth rate by an additional 0.3%, of which one-third, or 0.1%, can be attributed to China alone. In a more direct way, Feldkircher and Korhonen (2014) examine the global impact of an output shock from China. Using a global vector autoregressive model based on data of 43 economies from 1995Q1 to 2011Q4, they show that 1% positive output shock to China leads to a permanent increase of 1.1% in output of China and 0.1–0.5% in output of most large economies. Although this effect is non-negligible, it is quantitatively smaller than that of the United States. Their results show that 1% positive output shock to the United States leads to 0.6–1% increase in output of other advanced economies.

The world is affected not only by China’s trade, but also by its productivity growth, which refers to the spillover of one country’s productivity growth on other countries. Studies on international trade theory have identified two channels of these spillovers: the terms-of-trade effect and the home-market effect. Using data from 1995 to 2007, Hsieh and Ossa (2011) examine the global influence of China’s productivity growth through international trade and conclude that the cumulative welfare effect, through both channels, is 0.29% on average for the rest of the entire world. Di Giovanni *et al.* (2014) examine the global welfare effect of both China’s integration into the world and its productivity growth using a Ricardian–Heckscher–Ohlin model that is based on calibrated values for the years 2000 to 2007. They examine two cases, namely, the ‘balanced growth’ case, where every sector grows at the same rate, and the ‘unbalanced growth’ case where growth is biased toward China’s comparative advantage sectors. The results for productivity growth suggest that the mean global welfare gain in the unbalanced growth case is 0.42%, which is significantly larger than that in the balanced growth case, because China’s current comparative advantage pattern is similar to the world average and the unbalanced case has differentiated China further from the other countries. By considering the pattern of productivity growth, the estimation of Giovanni *et al.* is larger than that of Hsieh and Ossa (2011), who estimate the mean global welfare gain as 0.29%. For trade integration, Di Giovanni *et al.* (2014) compare the effect of China’s present trade integration level with the

hypothetical autarky case and reveal that the world benefits from China's trade liberalization. The mean global welfare gain is 0.13%, which is positive yet small. This finding is consistent with those of previous studies on China's trade integration.

In summary, the overall global influence of the GDP growth and trade integration of China is positive yet quantitatively small. The productivity growth of China is high and its spillover effect on the world is as large as that of trade integration.

5.2. *Distributional effects across regions*

Although the overall effect of the rise of China on the world is positive, each region is affected differently. Ianchovichina and Martin (2004) point out that most developed countries benefit from China's entry to the WTO, whereas some developing countries that compete with China in third markets may suffer from small losses. Among all countries and regions, China itself benefits the most (US\$40 552m), followed by Western European economies (US\$18 189m), North America (US\$6072m), Japan (US\$5694m) and Taiwan (US\$2985m). By contrast, India suffers the most (US\$3357m), followed by South Asian countries (US\$1622m). This finding is consistent with that of the IMF (2004), which indicates that most regions, except for South Asia, benefit from the output and trade growth of China. The loss of South Asia may be attributed to its similarities with China in its comparative advantages in labour-intensive sectors as well as its limited supply of intermediate goods to be exported to China. However, other parts of Asia, particularly East Asia, benefit from China's growth and integration. Di Giovanni *et al.* (2014) assert that, among all other regions, East and South Asian economies benefit the most (0.32%), whereas SSA and LAC countries benefit the least (0.08 and 0.09%, respectively). Only 9 out of 75 countries experience welfare losses from China's gains, which has also been confirmed by Levchenko and Zhang (2012), who argue that East and South Asia gain the most (0.7%), whereas Latin America and the Caribbean gain the least (0.16%) from the trade integration of China, India, and Central and Eastern Europe.

With regard to China's productivity growth, the cumulative welfare effect on individual regions ranges between -1.2% (for Russia) and 3.6% (for Asia except Japan). Only 3% of the total worldwide welfare gains from China's productivity growth accrue to the rest of the world, whereas the other 97% accrue to China. In particular, the effect is 0.4% for the United States (Hsieh and Ossa, 2011). Di Giovanni *et al.* (2014) point out that East and South Asia gain the most in both the balanced (0.03%) and unbalanced growth cases (0.84%).

Using various measures of competition, such as market share, complimentary index, competition index, and RCA, Qureshi and Wan (2008) also present some evidence on the distributional effects of China's export surge on different regions of the world. Countries/regions that face tough competition from China include some developed countries, such as the United States, advanced economies in East Asia, some European countries and some developing

countries, including Mexico.¹¹ However, countries from other regions of the world, such as Africa and Latin America, are not seriously affected by China's export surge.

5.3. *Distributional effects across sectors*

Labour-intensive manufacturings, such as the textiles, clothing, and apparel industries in which China has comparative advantages are negatively affected the most by China's growth (IMF, 2004; Ghosh and Rao, 2010; Levchenko and Zhang, 2012). Levchenko and Zhang (2012) simulations show that the trade integration of China, India, and Central and Eastern Europe to the world market will trigger absolute welfare losses (from 1.5 and 3.5%) for the leather and footwear, wearing apparel, furniture and textile industries. Similarly, Ghosh and Rao (2010) indicate that Canada, the United States and Europe all suffer losses in their textile and apparel industries upon the removal of the MFA quota. Specifically, the output and employment in Canada and United States have decreased by approximately 20 to 30%. However, the employment and output gains in their other industries slightly outnumber the losses in these two industries, which indicates that China's growth has a modest yet positive effect on the overall economy of these two countries.

Although competition from China is mainly concentrated in labour-intensive products, Qureshi and Wan (2008) also observe some competition in skill-intensive or middle-technology and high-technology products, as evidenced by China's increasing export shares in these products. Dimaranan *et al.* (2007) also suggest that the future efficiency improvements in the high-technology and heavy industries of China and India can bring severe competition in these sectors and crowd out other countries' market share in the global market, but leave opportunities for the light manufactures in these countries.

6. CONCLUSIONS

In this paper, we have reviewed research on China's global influence. The review gives a clear picture of the importance of China's integration into the global market. It highlights the regions and industries that receive positive or negative impacts. In most cases, quantitative aspects of the impacts are specified. Issues that are left untouched but are also of importance in this literature, will be for future research.

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¹¹ The European countries include Austria, Denmark, France, Germany, Italy, the Netherlands, Portugal, Spain, and the UK.

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